Configuração 8: M = 128, C = 8, K = 12, IoT = 20 dB, QPSK

Perda de SNR do BCD-MMSE (iter 1) em SER = 10^{-2}, em dB, em relação à referência ideal (float64); × = não chega a 10^{-2} nas SNRs simuladas

Fundo laranja no rótulo: o modo com o melhor resultado (estouro: pela varredura de I; quantização: pela varredura de F), comparando de menos bits para mais, sem contar a coluna em que tudo é zero

Faixa "x a y": o outro modo muda o valor (a quantização no mapa de I, o estouro no mapa de F); se os dois extremos têm cores diferentes, a célula é dividida na diagonal (em cima, a cor do menor; embaixo, a do maior)

Embaixo da perda: maior ocupação da XCU50 entre LUT, DSP, FF e BRAM (síntese HLS, sem diretivas), no mesmo formato e modo; sem número onde o formato e o modo não foram sintetizados

TDLA30-5: N = 192, f_c = 3,5 GHz

Estouro	6	8	10	12	14
AP_SAT	× 5.5 a 6.6%	× 6.0 a 7.2%	× 6.1 a 7.2%	0.33 6.1 a 7.3%	0.00 6.2 a 7.3%
AP_SAT_ZERO	× 6.6%	× 7.1 a 7.2%	× 7.1 a 7.2%	0.27 7.2 a 7.3%	0.00 7.7 a 7.8%
AP_SAT_SYM	× 6.5 a 6.6%	× 7.1 a 7.2%	× 7.1 a 7.2%	0.33 7.2 a 7.3%	0.00 7.2 a 7.3%
AP_WRAP	× 5.8 a 6.6%	× 6.3 a 7.2%	4.56 6.4 a 7.2%	0.17 6.4 a 7.3%	0.00 6.9 a 7.8%
AP_WRAP_SM	× 6.6%	× 7.1 a 7.2%	× 7.1 a 7.2%	0.38 7.2 a 7.3%	0.00 7.2 a 7.3%

Bits inteiros I (F = 32)

TDLA30-5: N = 192, f_c = 3,5 GHz

Quantização	4	6	8	10	12	14
AP_TRN	×	×	× 1.5 a 1.7%	× 1.7 a 1.8%	1.46 1.9 a 2.1%	0.04 2.0 a 2.2%
AP_TRN_ZERO	×	×	× 1.4 a 2.0%	× 1.6 a 2.1%	1.28 1.9 a 2.3%	0.00 2.0 a 2.5%
AP_RND	×	×	× 1.6 a 1.9%	2.77 1.7 a 2.1%	0.16 2.0 a 2.3%	0.01 2.1 a 2.4%
AP_RND_ZERO	×	×	× 1.4 a 2.0%	2.78 1.6 a 2.1%	0.16 1.9 a 2.3%	0.01 2.0 a 2.5%
AP_RND_MIN_INF	×	×	× 1.6 a 1.9%	2.80 1.8 a 2.1%	0.16 2.1 a 2.3%	0.00 2.2 a 2.4%
AP_RND_INF	×	×	× 1.4 a 2.0%	2.79 1.6 a 2.1%	0.16 1.9 a 2.3%	0.00 2.0 a 2.5%
AP_RND_CONV	×	×	× 1.6 a 2.0%	2.83 1.8 a 2.1%	0.17 2.1 a 2.3%	0.01 2.2 a 2.5%

Bits fracionários F (I = 14)

TDLA30-75: N = 192, f_c = 28 GHz

Estouro	6	8	10	12	14
AP_SAT	× 5.5 a 6.6%	× 6.0 a 7.2%	× 6.1 a 7.2%	0.30 6.1 a 7.3%	0.00 6.2 a 7.3%
AP_SAT_ZERO	× 6.6%	× 7.1 a 7.2%	× 7.1 a 7.2%	0.24 7.2 a 7.3%	0.00 7.7 a 7.8%
AP_SAT_SYM	× 6.5 a 6.6%	× 7.1 a 7.2%	× 7.1 a 7.2%	0.30 7.2 a 7.3%	0.00 7.2 a 7.3%
AP_WRAP	× 5.8 a 6.6%	× 6.3 a 7.2%	4.06 a 4.5 6.4 a 7.2%	0.00 6.4 a 7.3%	0.00 6.9 a 7.8%
AP_WRAP_SM	× 6.6%	× 7.1 a 7.2%	× 7.1 a 7.2%	0.31 7.2 a 7.3%	0.00 7.2 a 7.3%

Bits inteiros I (F = 32)

TDLA30-75: N = 192, f_c = 28 GHz

Quantização	4	6	8	10	12	14	16
AP_TRN	×	×	× 1.5 a 1.7%	× 1.7 a 1.8%	1.53 1.9 a 2.1%	0.06 2.0 a 2.2%	0.01 3.1 a 3.7%
AP_TRN_ZERO	×	×	× 1.4 a 2.0%	× 1.6 a 2.1%	1.18 1.9 a 2.3%	0.01 2.0 a 2.5%	0.00 3.2 a 3.7%
AP_RND	×	×	× 1.6 a 1.9%	2.74 1.7 a 2.1%	0.19 2.0 a 2.3%	0.02 2.1 a 2.4%	0.00 3.5 a 3.7%
AP_RND_ZERO	×	×	× 1.4 a 2.0%	2.73 1.6 a 2.1%	0.19 1.9 a 2.3%	0.02 2.0 a 2.5%	0.00 3.2 a 3.7%
AP_RND_MIN_INF	×	×	× 1.6 a 1.9%	2.75 1.8 a 2.1%	0.19 2.1 a 2.3%	0.02 2.2 a 2.4%	0.00 3.7%
AP_RND_INF	×	×	× 1.4 a 2.0%	2.76 1.6 a 2.1%	0.20 1.9 a 2.3%	0.02 2.0 a 2.5%	0.00 3.2 a 3.7%
AP_RND_CONV	×	×	× 1.6 a 2.0%	2.76 1.8 a 2.1%	0.19 2.1 a 2.3%	0.02 2.2 a 2.5%	0.00 3.7%

Bits fracionários F (I = 14)

TDLA30-300: N = 168, f_c = 28 GHz

Estouro	6	8	10	12	14
AP_SAT	× 5.5 a 6.6%	× 6.0 a 7.2%	× 6.1 a 7.2%	0.38 6.1 a 7.3%	0.00 6.2 a 7.3%
AP_SAT_ZERO	× 6.6%	× 7.1 a 7.2%	× 7.1 a 7.2%	0.23 7.2 a 7.3%	0.00 7.7 a 7.8%
AP_SAT_SYM	× 6.5 a 6.6%	× 7.1 a 7.2%	× 7.1 a 7.2%	0.38 7.2 a 7.3%	0.00 7.2 a 7.3%
AP_WRAP	× 5.8 a 6.6%	× 6.3 a 7.2%	3.81 a 4.4 6.4 a 7.2%	0.09 a 0.10 6.4 a 7.3%	0.00 6.9 a 7.8%
AP_WRAP_SM	× 6.6%	× 7.1 a 7.2%	× 7.1 a 7.2%	0.33 7.2 a 7.3%	0.00 7.2 a 7.3%

Bits inteiros I (F = 32)

TDLA30-300: N = 168, f_c = 28 GHz

Quantização	4	6	8	10	12	14	16
AP_TRN	×	×	× 1.5 a 1.7%	× 1.7 a 1.8%	1.51 1.9 a 2.1%	0.06 2.0 a 2.2%	0.00 3.1 a 3.7%
AP_TRN_ZERO	×	×	× 1.4 a 2.0%	× 1.6 a 2.1%	1.39 1.9 a 2.3%	0.02 2.0 a 2.5%	0.00 3.2 a 3.7%
AP_RND	×	×	× 1.6 a 1.9%	2.87 1.7 a 2.1%	0.19 2.0 a 2.3%	0.02 2.1 a 2.4%	0.00 3.5 a 3.7%
AP_RND_ZERO	×	×	× 1.4 a 2.0%	2.87 1.6 a 2.1%	0.19 1.9 a 2.3%	0.02 2.0 a 2.5%	0.00 3.2 a 3.7%
AP_RND_MIN_INF	×	×	× 1.6 a 1.9%	2.87 1.8 a 2.1%	0.19 2.1 a 2.3%	0.02 2.2 a 2.4%	0.00 3.7%
AP_RND_INF	×	×	× 1.4 a 2.0%	2.87 1.6 a 2.1%	0.19 1.9 a 2.3%	0.02 2.0 a 2.5%	0.00 3.2 a 3.7%
AP_RND_CONV	×	×	× 1.6 a 2.0%	2.87 1.8 a 2.1%	0.19 2.1 a 2.3%	0.02 2.2 a 2.5%	0.00 3.7%

Bits fracionários F (I = 14)

TDLC300-100: N = 132, f_c = 3,5 GHz

Estouro	6	8	10	12	14
AP_SAT	× 5.5 a 6.6%	× 6.0 a 7.2%	× 6.1 a 7.2%	0.30 6.1 a 7.3%	0.00 6.2 a 7.3%
AP_SAT_ZERO	× 6.6%	× 7.1 a 7.2%	× 7.1 a 7.2%	0.20 a 0.22 7.2 a 7.3%	0.00 7.7 a 7.8%
AP_SAT_SYM	× 6.5 a 6.6%	× 7.1 a 7.2%	× 7.1 a 7.2%	0.30 7.2 a 7.3%	0.00 7.2 a 7.3%
AP_WRAP	× 5.8 a 6.6%	× 6.3 a 7.2%	4.48 a 4.5 6.4 a 7.2%	0.09 6.4 a 7.3%	0.00 6.9 a 7.8%
AP_WRAP_SM	× 6.6%	× 7.1 a 7.2%	× 7.1 a 7.2%	0.32 7.2 a 7.3%	0.00 7.2 a 7.3%

Bits inteiros I (F = 32)

TDLC300-100: N = 132, f_c = 3,5 GHz

Quantização	4	6	8	10	12	14	16
AP_TRN	×	×	× 1.5 a 1.7%	× 1.7 a 1.8%	1.56 1.9 a 2.1%	0.07 2.0 a 2.2%	0.01 3.1 a 3.7%
AP_TRN_ZERO	×	×	× 1.4 a 2.0%	× 1.6 a 2.1%	1.58 1.9 a 2.3%	0.05 2.0 a 2.5%	0.00 3.2 a 3.7%
AP_RND	×	×	× 1.6 a 1.9%	2.92 1.7 a 2.1%	0.25 2.0 a 2.3%	0.01 2.1 a 2.4%	0.00 3.5 a 3.7%
AP_RND_ZERO	×	×	× 1.4 a 2.0%	2.92 1.6 a 2.1%	0.25 1.9 a 2.3%	0.01 2.0 a 2.5%	0.00 3.2 a 3.7%
AP_RND_MIN_INF	×	×	× 1.6 a 1.9%	2.90 1.8 a 2.1%	0.26 2.1 a 2.3%	0.01 2.2 a 2.4%	0.00 3.7%
AP_RND_INF	×	×	× 1.4 a 2.0%	2.91 1.6 a 2.1%	0.26 1.9 a 2.3%	0.01 2.0 a 2.5%	0.00 3.2 a 3.7%
AP_RND_CONV	×	×	× 1.6 a 2.0%	2.90 1.8 a 2.1%	0.25 2.1 a 2.3%	0.01 2.2 a 2.5%	0.00 3.7%

Bits fracionários F (I = 14)

TDLB100-400: N = 24, f_c = 3,5 GHz

Estouro	6	8	10	12
AP_SAT	× 5.5 a 6.6%	× 6.0 a 7.2%	2.80 6.1 a 7.2%	0.00 6.1 a 7.3%
AP_SAT_ZERO	× 6.6%	× 7.1 a 7.2%	2.83 a 2.8 7.1 a 7.2%	0.00 7.2 a 7.3%
AP_SAT_SYM	× 6.5 a 6.6%	× 7.1 a 7.2%	2.80 7.1 a 7.2%	0.00 7.2 a 7.3%
AP_WRAP	× 5.8 a 6.6%	× 6.3 a 7.2%	2.86 a 2.9 6.4 a 7.2%	0.00 6.4 a 7.3%
AP_WRAP_SM	× 6.6%	× 7.1 a 7.2%	3.07 7.1 a 7.2%	0.00 7.2 a 7.3%

Bits inteiros I (F = 32)

TDLB100-400: N = 24, f_c = 3,5 GHz

Quantização	4	6	8	10	12	14	16	20
AP_TRN	×	×	× 1.5 a 1.7%	5.84 1.7 a 1.8%	1.15 1.9 a 2.1%	0.04 2.0 a 2.2%	0.00 3.1 a 3.7%	0.00 3.6 a 4.2%
AP_TRN_ZERO	×	×	× 1.4 a 2.0%	× 1.6 a 2.1%	2.47 1.9 a 2.3%	0.45 2.0 a 2.5%	0.07 3.2 a 3.7%	0.00 3.7 a 4.2%
AP_RND	×	×	× 1.6 a 1.9%	2.50 1.7 a 2.1%	0.35 2.0 a 2.3%	0.02 2.1 a 2.4%	0.00 3.5 a 3.7%	0.00 4.1 a 4.2%
AP_RND_ZERO	×	×	× 1.4 a 2.0%	2.51 1.6 a 2.1%	0.35 1.9 a 2.3%	0.02 2.0 a 2.5%	0.00 3.2 a 3.7%	0.00 3.7 a 4.2%
AP_RND_MIN_INF	×	×	× 1.6 a 1.9%	2.51 1.8 a 2.1%	0.36 2.1 a 2.3%	0.02 2.2 a 2.4%	0.00 3.7%	0.00 4.2%
AP_RND_INF	×	×	× 1.4 a 2.0%	2.51 1.6 a 2.1%	0.35 1.9 a 2.3%	0.02 2.0 a 2.5%	0.00 3.2 a 3.7%	0.00 3.7 a 4.2%
AP_RND_CONV	×	×	× 1.6 a 2.0%	2.51 1.8 a 2.1%	0.35 2.1 a 2.3%	0.02 2.2 a 2.5%	0.00 3.7%	0.00 4.2%

Bits fracionários F (I = 14)